

CHAPTER 7: HUMAN HEALTH AND DISEASE • FAST-TRACK REVISION NOTES

PART 1 • CORE ETIOLOGY & PATHOGENS

1. HEALTH DEFINITION & PILLARS OF DISEASE RESISTANCE

DEFINITION & MAINTENANCE OF HEALTH

Health (WHO): State of complete *physical, mental, and social well-being*, not merely absence of disease. World Health Day: **April 7**.

- **Pillars:** Balanced diet, personal hygiene, exercise/yoga, vaccination, clean water, proper sewage/waste disposal & vector eradication.

ETIOLOGY OF HUMAN DISEASES

- **1. Genetic factors:** Congenital defects, inborn metabolic errors.
- **2. Infectious agents:** Invasion by biological pathogens.
- **3. Immunological failures:** Inborn/acquired immunodeficiency.
- **4. Lifestyle:** Sedentary habits, stress, junk food, substance abuse.

2. PATHOGENIC DISEASES IN HUMANS: HIGH-YIELD REVISION MATRIX

Disease & Class	Causative Pathogen	Vector / Route	Target Tissue	Key Pathognomonic Symptoms & Exam Pointers
Typhoid Bacterial	<i>Salmonella typhi</i> (Gram-ve bacillus)	Fecal-oral route (contaminated food/ water)	Small intestine → enters blood	Sustained high fever (39°–40°C), stomach pain, constipation, headache, anorexia. Severe: intestinal perforation . Carrier: <i>Mary Mallon (Typhoid Mary)</i> . Confirm: Widal Test.
Pneumonia Bacterial	<i>Streptococcus pneumoniae</i> , <i>H. influenzae</i>	Inhalation of droplets/ aerosols; shared cups	Alveoli of lungs (exudate accumulates)	Fever, severe chills, productive cough, dyspnea. Severe hypoxia: lips and fingernails turn slate-grey to cyanotic bluish .
Common Cold Viral	Rhino viruses (>100 serotypes)	Inhaled droplets; fomites (pens, cups, keyboard)	Nose & upper respiratory tract (NOT lungs)	Nasal congestion, profuse rhinorrhea, sore throat, hoarseness, cough, headache; typically resolves naturally within 3–7 days.
Malaria Protozoan	<i>Plasmodium</i> (<i>vivax</i> , <i>falciparum</i> , <i>malariae</i>)	Bite of female <i>Anopheles</i> mosquito	Liver hepatocytes & RBCs	Rupture of RBCs releases hemozoin → chills & recurring high fever. <i>P. falciparum</i> induces lethal malignant/ cerebral malaria.
Amoebiasis Protozoan	<i>Entamoeba histolytica</i>	Houseflies (mechanical vector); food/water	Mucosa/submucosa of large intestine	Abdominal cramps, constipation, stools containing excessive mucous and dark blood clots.
Ringworm Fungal	<i>Microsporum</i> , <i>Trichophyton</i> , <i>Epidermophyton</i>	Direct contact, towels, combs, clothes, soil	Keratinized tissue: Skin, scalp, nails, groin folds	Dry, circumscribed scaly lesions with intense itching (pruritus). Heat and moisture promote fungal hyphal growth.
Ascariasis Helminthic	<i>Ascaris lumbricoides</i> (Roundworm)	Water, unwashed fruits/veggies contaminated by feces	Lumen of small intestine	Internal hemorrhages, muscular pain, persistent low fever, severe anemia, mechanical obstruction of intestinal passage.
Elephantiasis (Filariasis) Helminth	<i>Wuchereria bancrofti</i> , <i>Wuchereria malayi</i>	Bite of infective female <i>Culex</i> mosquito	Lymphatic vessels of lower limbs & genitalia	Chronic granulomatous inflammation of lymphatic endothelium; grotesque swelling & lymphedema of lower extremities/scrotum.

3. LIFE CYCLE OF PLASMODIUM (DIGENETIC PARASITE: HUMAN + MOSQUITO)

HUMAN HOST (INTERMEDIATE • ASEXUAL SCHIZOGONY)

- **1. Inoculation:** Mosquito bite injects infectious spindle-shaped **sporozoites** into blood.
- **2. Hepatic Schizogony:** Sporozoites reach liver hepatocytes, multiply asexually → burst liver cells → release **merozoites** into blood.
- **3. Erythrocytic Schizogony:** Merozoites invade RBCs → multiply asexually → mature schizonts burst RBCs.
- **4. Toxic Release:** Rupture discharges **hemozoin** granules, initiating bouts of chills and spiking fever every 3–4 days.
- **5. Gametocytogenesis:** A fraction of merozoites differentiate into ♂ microgametocytes and ♀ macrogametocytes in RBCs.

MOSQUITO HOST (DEFINITIVE • SEXUAL SPOROGONY)

- **1. Ingestion:** Female *Anopheles* takes blood meal from infected human, ingesting mature ♂ and ♀ gametocytes.
- **2. Syngamy:** In **mosquito midgut** (stomach), microgametes fertilize macrogametes → forms motile **ookinete**.
- **3. Penetration:** Ookinete bores through the stomach wall and encysts on outer surface as an **oocyst**.
- **4. Sporogony:** Oocyst undergoes nuclear fission, producing thousands of infective **sporozoites**.
- **5. Migration:** Oocyst bursts → sporozoites travel through hemolymph to lodge in **salivary glands** for next human transmission.

4. INNATE IMMUNITY: FOUR NON-SPECIFIC BARRIERS

Barrier Classification	Anatomical / Chemical Components	Mechanism of Action & Role
1. Physical Barriers	Keratinized stratum corneum of skin; Mucus coating of respiratory, gastrointestinal, and urogenital tracts	Mechanical barrier blocks microbe entry; sticky mucus traps foreign pathogens and cilia sweep them upward.
2. Physiological Barriers	Gastric HCl (pH 1.5–2.5), lysozyme in tears & saliva, bile salts, skin sebum (acid mantle)	Acidic gastric juice kills swallowed microbes; lysozyme hydrolyzes bacterial peptidoglycan cell walls.
3. Cellular Barriers	Polymorphonuclear leukocytes (PMNL-neutrophils), Monocytes, tissue Macrophages, Natural Killer (NK) cells	Phagocytose and degrade invading microorganisms; NK cells induce apoptosis in virus-infected and cancerous cells.
4. Cytokine Barriers	Interferons (α , β , γ glycoproteins)	Secreted by virus-infected cells; bind uninfected neighbors to induce antiviral proteins, aborting viral replication.

5. ACQUIRED IMMUNITY, ANTIBODY ARCHITECTURE & EFFECTOR ARMS

ANTIBODY STRUCTURE (H_2L_2) & IG CLASSES

Formulated as H_2L_2 : 2 Heavy chains (~50 kDa) + 2 Light chains (~25 kDa) joined by covalent **disulfide (S–S) bonds**.

- **Variable (V) Region (F_{ab})**: Tips of Y; forms antigen-binding site (**paratope**) specific to the epitope.
- **Constant (C) Region (F_c)**: Stem of Y; mediates macrophage binding and complement activation.
- **IgG**: Most abundant (~75–80%); *only antibody crossing placenta*.
- **IgA**: Secretory dimer; abundant in **colostrum** (mother's first milk).
- **IgM**: Pentamer; produced first during primary immune response.
- **IgE**: Mediates type-I hypersensitivity (allergic response).

HIR VS. CMIR & GRAFT REJECTION

- **Humoral Immune Response (HIR)**:
 - Mediated by circulating **B-lymphocytes**.
 - B-cells differentiate into antibody-secreting **plasma cells** and long-lived **memory B-cells**.
 - Antibodies in humors (blood/lymph) neutralize pathogens.
- **Cell-Mediated Immune Response (CMIR)**:
 - Mediated directly by **T-lymphocytes** (CD4+ Helper & CD8+ Cytotoxic T-cells).
 - Attacks intracellular pathogens, viruses, and tumor cells.
- **Graft Rejection**: T-cells distinguish 'self' from 'non-self' tissue antigens, causing transplant rejection (requires HLA tissue matching & immunosuppressants like Cyclosporin A).

6. ACTIVE VS. PASSIVE IMMUNITY: COMPARATIVE MATRIX

Criterion	Active Immunity	Passive Immunity
Antibody Source	Produced endogenously by the individual's own B-lymphocytes	Exogenous readymade antibodies received from another organism
Latency & Efficacy	Slow onset; requires a distinct lag period for clonal expansion	Immediate physiological protection; instantly neutralizes toxins
Duration & Memory	Long-lasting to permanent; establishes immunological memory	Transient & short-lived; foreign immunoglobulins are catabolized
Natural Example	Natural recovery following chickenpox, measles, or typhoid infection	Maternal IgG across placenta; secretory IgA via colostrum to infant
Artificial Example	Prophylactic vaccines (Polio OPV, BCG, MMR, Hepatitis B)	Anti-tetanus serum (ATS); Anti-snake venom (ASV) injections

7. LYMPHOID ORGANS & IMMUNOPATHOLOGY (ALLERGY & AUTOIMMUNITY)

LYMPHOID ORGANS: CENTRAL VS. PERIPHERAL

- **Primary (Central)**: Sites of lymphocyte genesis & maturation:
 - **Bone Marrow**: Produces all blood cells; B-cells mature here.
 - **Thymus**: T-cells mature here; large at birth, shrinks with age.
- **Secondary (Peripheral)**: Sites of antigen trapping & proliferation:
 - **Spleen**: Blood filter, RBC graveyard, erythrocyte reservoir.
 - **Lymph Nodes**: Filter tissue lymph, trap antigens, activate T/B cells.
 - **MALT**: Lines digestive, respiratory, urogenital tracts; = **50%** of human lymphoid tissue.

ALLERGIES & AUTOIMMUNITY

- **Allergy (Hypersensitivity)**: Exaggerated response to harmless allergens (pollen, dust, animal dander).
 - Mediated by **IgE antibodies**.
 - IgE cross-links on **mast cells & basophils** → releases **histamine** and **serotonin**.
 - Relieved by *antihistamines, adrenaline, corticosteroids*.
- **Autoimmunity**: Loss of self-tolerance; body attacks self-antigens.
 - Landmark: **Rheumatoid Arthritis** — autoantibodies attack joint synovial membranes → severe deformity and pain.
 - Others: Myasthenia gravis, Type-1 diabetes, Vitiligo.

8. AIDS (HIV REPLICATION) & ONCOLOGY (CANCER BIOLOGY)

AIDS: RETROVIRAL CASCADE & CD4 DEPLETION

- **Agent:** HIV (Retrovirus, envelope with gp120 spikes, 2 ssRNA + reverse transcriptase). Latency: 5–10 yrs.
- **Transmission:** Unprotected sex, contaminated blood, shared IV needles, transplacental (mother → child).
- **Pathogenesis:**
 - gp120 attaches to **CD4 receptors** on macrophages & T_H cells.
 - Reverse transcriptase transcribes viral RNA → viral DNA.
 - Integrase integrates viral DNA into host genome.
 - *Macrophages survive and act as an continuous HIV factory.*
 - HIV attacks & destroys **CD4+ Helper T-cells (T_H)** → T_H count plunges (<200/μL) → fatal opportunistic infections (*Mycobacterium*, *Toxoplasma*, fungi, viruses).
- **Diagnosis:** **ELISA** (screening); **Western Blot** (confirmatory).
- **Treatment:** cART (e.g., **AZT / Zidovudine** — reverse transcriptase inhibitor). Controlled by **NACO** and NGOs.

CANCER BIOLOGY: PROPERTIES, CAUSES & THERAPY

- **Hallmark:** Loss of **contact inhibition** → uncontrolled division forming **tumors**.
- **Tumor Types:**
 - **Benign:** Confined to primary site, encapsulated, non-fatal.
 - **Malignant:** Infiltrative; shows **metastasis** (cells slough off and seed secondary tumors at distant organs — fatal property).
- **Carcinogens:**
 - **Physical:** Ionizing (X-rays, γ-rays) & Non-ionizing (UV).
 - **Chemical:** Tobacco smoke mutagens (lung/bladder cancer).
 - **Biological:** Oncogenic viruses, proto-oncogene activation (c-onc).
- **Diagnosis:** Biopsy/histopathology, Radiography, CT scan (3D X-rays), MRI (magnetic resonance), monoclonal antibody tests.
- **Therapy:** Surgery, Radiotherapy, Chemotherapy (causes alopecia, anemia), and **Immunotherapy** (biological response modifiers like **α-interferon** activate immune tumor destruction).

9. ABUSED DRUGS, SOURCES, RECEPTOR TARGETS & CLINICAL ACTIONS

Drug Class	Source / Plant Origin	Receptor Target & Mechanism	Physiological Effects & Clinical Remarks
Opioids (Heroin / Smack, Morphine)	Latex of Opium Poppy (<i>Papaver somniferum</i>). Heroin = diacetylmorphine (acetylation of morphine). Bitter, odorless white solid.	Binds specific opioid receptors in the CNS and Gastrointestinal tract .	Potent depressant ; slows down body functions; powerful pain killer (analgesic). Consumed by snorting or injection. Overdose leads to respiratory failure.
Cannabinoids (Marijuana, Hashish, Charas, Ganja)	Inflorescences, flower tops, leaves, and resins of Indian hemp (<i>Cannabis sativa</i>). Inhaled or ingested.	Interacts with cannabinoid receptors concentrated in the brain neocortex .	Alters sensory perception; impairs motor coordination and cognitive timing; markedly alters the cardiovascular system (tachycardia). Abused by athletes.
Coca Alkaloids (Cocaine, Coke, Crack)	Leaves of South American coca bush (<i>Erythroxylum coca</i>). Usually snorted.	Blocks the reuptake mechanism of neurotransmitter Dopamine at synaptic junctions.	Potent CNS stimulant ; induces euphoria, hyperactivity, and increased energy. High doses cause paranoid delusions and intense hallucinations .
Hallucinogens	Botanical extracts from <i>Atropa belladonna</i> and <i>Datura</i> ; Fungus derivative: LSD .	Disrupts serotonergic neurotransmission in cerebral cortical centers.	Produces vivid visual/auditory illusions, acute psychosis, and frightening panic episodes ("bad trips").
Prescription Misuse	Barbiturates, Amphetamines, Benzodiazepines, Morphine.	Prescribed medically for insomnia, anxiety, and post-surgical trauma.	High dependence liability; abused recreationally. Morphine is a clinically indispensable surgical analgesic.
Tobacco & Nicotine	Dried leaves of <i>Nicotiana tabacum</i> (smoked, chewed, or snuffed).	Stimulates adrenal medulla to secrete adrenaline & noradrenaline .	Elevates heart rate and blood pressure. Smoking causes bronchitis, emphysema, oral/lung cancer; increases blood CO, decreasing oxyhemoglobin.

10. ADDICTION DYNAMICS, STEROID HAZARDS & PREVENTION GUIDELINES

ADDICTION, DEPENDENCE & WITHDRAWAL

- **Addiction:** Psychological attachment to euphoric effects.
- **Tolerance:** Receptors down-regulate → progressively higher doses required to elicit response.
- **Dependence & Withdrawal:** Body adapts to drug; sudden stoppage triggers **Withdrawal Syndrome**: anxiety, tremors, sweating, nausea, tachycardia.
- **Systemic Harm:** IV needles transmit **HIV & Hepatitis B**; chronic alcohol causes lethal **Liver Cirrhosis**.

ADVERSE EFFECTS OF ANABOLIC STEROIDS

- **In Females:** Masculinization, hirsutism (excess facial hair), aggressiveness, menstrual irregularities, clitoromegaly, deepening of voice.
- **In Males:** Severe acne, aggressiveness, depression, testicular atrophy, oligospermia (reduced sperm), gynecomastia (breast development), premature baldness, enlarged prostate.

FIVE PILLARS FOR PREVENTION AND CONTROL OF DRUG/ALCOHOL ABUSE

- 1. Avoid Peer Pressure:** Respect unique individual limits & interests.
- 2. Education & Counseling:** Impart resilience to accept failure as normal.
- 3. Family Support:** Foster open communication without guilt/fear.
- 4. Danger Signals:** Identify sudden drops in academic focus, isolation, sleep shifts.
- 5. Medical Help:** Seek de-addiction centers & cognitive behavioral therapy.